

Prasad Pramod Patil

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SUMMARY

Data Engineer with 2+ years building production-grade ETL pipelines, Snowflake data warehouses, and real-time streaming systems using Python, SQL, PySpark, dbt, Kafka, and Airflow – delivering reliable analytics-ready datasets across healthcare, retail, supply chain, and financial domains.

EXPERIENCE

System Engineer – Data Pipeline Engineering and ETL Development July 2022 – July 2024
Tata Consultancy Services, Mumbai, India

- Processed 450M+ records using Oracle and SQL Server building scalable ETL pipelines integrating enterprise data sources.
- Refactored 200+ SQL Server stored procedures optimizing query performance and reducing pipeline execution failures by 30%.
- Engineered automated Python data workflows replacing manual ETL processes cutting data delivery turnaround by 50%.
- Diagnosed 85+ pipeline integrity issues implementing validation logic reducing downstream analytical failures by 40%.

ADDITIONAL EXPERIENCE

Data Analytics and Pipeline Engineer – ETL Automation and Data Quality August 2025 – December 2025
JMA Wireless, Syracuse, NY

- Constructed SQL and Python ETL pipelines processing 100K+ operational records ensuring quality across transformation stages.
- Identified 47.75% process failure rate implementing data quality checks reducing pipeline failures and SLA breaches by 40%.
- Developed Tableau and Power BI dashboards communicating KPI trends and pipeline health to senior engineering stakeholders.
- Applied statistical validation confirming transformation accuracy and documenting technical specifications for data pipelines.

TECHNICAL SKILLS

- **Languages:** Python, PySpark, Spark SQL, SQL, R
- **Data Platforms:** Snowflake, Databricks, dbt, Delta Lake, Medallion Architecture, Star Schema
- **Orchestration and Cloud:** Apache Airflow, Apache Kafka, AWS (S3, Glue, Lambda, CloudWatch), Docker, Git
- **Data Quality and Serving:** dbt Tests, Data Validation, Schema Enforcement, FastAPI, ETL/ELT Pipelines
- **Analytics and ML:** Power BI, Tableau, XGBoost, SHAP, Prophet, SARIMA, Random Forest, SMOTE

PROJECTS

SmartShop Analytics – Real-Time Kafka Streaming Pipeline and Delta Lake Warehouse – *GitHub*

- Architected Kafka event-driven ETL pipeline on Docker ingesting 100K+ events with exactly-once fault-tolerant delivery.
- Deployed PySpark Structured Streaming consumer persisting enriched events to Delta Lake with automated schema enforcement.
- Modeled 4 dbt models on Databricks with dbt tests surfacing \$15.42M revenue and 97.02% delivery KPIs across 27 states.

NYC Taxi Medallion ELT Pipeline – Snowflake Warehouse and Airflow Orchestration – *GitHub*

- Transformed 5.7M records using PySpark across Bronze, Silver, Gold medallion architecture on Snowflake data warehouse.
- Authored dbt models with dbt tests enforcing schema consistency and data governance across analytical warehouse layers.
- Orchestrated Airflow DAGs with CI/CD workflows automating scheduled ELT execution and resolving failures independently.

Supply Chain Intelligence – Star Schema ETL Pipeline and Demand Forecasting – *GitHub*

- Designed star schema data model and Python ETL pipeline ingesting 180,519 supply chain records across 23 global regions.
- Surfaced \$21M revenue at risk using advanced SQL CTEs and window functions supporting cross-functional business decisions.
- Constructed Prophet demand forecast achieving 0.6% MAPE automating recurring forecasting for operational planning teams.

EDUCATION

Syracuse University August 2024 – May 2026
Master of Science in Applied Data Science GPA: 3.9

University of Mumbai June 2018 – May 2022
Bachelor of Engineering in Information Technology GPA: 3.8